

# PARADIGMATIC ANALYSIS

## *"Influence of Deep Tech-Based Digital Transformation on Quality and Efficiency of University Education in Emerging Contexts"*

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### 1. ONTOLOGICAL DIMENSION

#### What kind of reality are we assuming?

Central question: Is "educational quality" and "university efficiency" an objective reality existing independently of the observer, or are they social constructions shaped by historical, cultural, and institutional contexts?

#### Ontological tension in the title:

The term "influence" presupposes a causal relationship between Deep Tech and educational quality, implicitly assuming both phenomena are stable and measurable entities. However:

- Educational quality is not an intrinsic property — the same university can be "high quality" by international rankings and "deficient" according to local communities that value cultural relevance over indexed publications.
- Digital transformation is not a uniform process — in emerging contexts, multiple technological realities coexist simultaneously: hyperconnected universities alongside institutions with intermittent connectivity.
- Deep Tech technologies (AI, blockchain, quantum computing, biotechnology) are not neutral — they embody particular ontologies about what it means to learn, know, and what counts as evidence of learning.

#### Recommended ontological position:

Adopt a critical realism — acknowledge that real structures exist (technological infrastructure, pedagogical practices, digital divides) but that their meaning and impact are mediated by social and institutional mechanisms not directly observable. This requires operationally defining:

- Quality = graduation rates? Employability? Critical thinking development? Student satisfaction?
- Efficiency = cost per student? Time to graduation? Retention?
- Emerging context = geography? Level of economic development? Institutional maturity?

Without these definitions, the research builds on sand — measuring shadows of never-clarified concepts.

### 2. EPISTEMOLOGICAL DIMENSION

#### How can we know this influence, and what counts as valid knowledge?

Central question: How do we know that Deep Tech technologies effectively transform educational quality, and not simply coat it with an appearance of modernization?

### **Epistemological tension in the title:**

Asserting "influence" implies the researcher can epistemically access a causal relationship between complex variables in dynamic educational systems. This faces at least three serious problems:

#### **Problem 1 — The causality vs. correlation problem:**

A university adopting generative AI may simultaneously improve its ranking, increase student dropout, and reduce depth of critical thinking. Which of these effects represents the 'real influence' of technology? Positivist epistemology would select measurable indicators; interpretive epistemology would ask what each means for the actors involved.

#### **Problem 2 — The problem of situated knowledge:**

In emerging contexts, knowledge about educational quality is asymmetrically distributed: university administrators have access to administrative data, teachers know real classroom practices, students experience learning subjectively, and communities evaluate the social relevance of graduates. No single source is epistemologically sufficient.

#### **Problem 3 — The problem of technology as a black box:**

Deep Tech technologies frequently operate through opaque algorithms — an AI system that personalizes learning paths makes decisions whose logic neither teachers nor students can inspect. Knowing the 'influence' of something that cannot be opened epistemologically is a simultaneous methodological and ethical challenge.

### **Recommended epistemological position:**

A post-positivist or moderate constructivist approach combining:

- Quantitative performance and efficiency indicators (what changes)
- Interpretive analysis of teaching and student experiences (how and why it changes)
- Triangulation of sources to compensate for the limitations of each epistemic perspective

## **3. METHODOLOGICAL DIMENSION**

### **What research design is coherent with the title's claims?**

Central question: Does the chosen methodological design have the real capacity to capture the influence of complex technologies on equally complex educational systems?

### **Methodological tension in the title:**

The title formulates a relationship of influence — which methodologically requires, in its most rigorous form, a design capable of isolating the effect of the technological variable on educational

outcomes, controlling for confounding factors. In emerging contexts this is extraordinarily difficult because:

**Problem 1 — Structurally limited samples:**

Emerging contexts frequently involve small universes of universities with real Deep Tech adoption. A sample of 3 to 8 institutions does not support generalizable statistical inference, but can generate deep and transferable understanding if the design is coherently qualitative.

**Problem 2 — Temporality of the phenomenon:**

Digital transformation is a process, not an event. A cross-sectional design (snapshot at one moment) captures state of technological adoption, not trajectory of transformation. To study real influence, longitudinal designs are needed that most graduate research cannot sustain.

**Problem 3 — Heterogeneity of the Deep Tech construct:**

AI, blockchain, augmented reality, and quantum computing have radically different mechanisms of influence on education. Treating them as a unified category produces methodologically imprecise findings — it is like studying 'the influence of transportation' without distinguishing between bicycle, airplane, and train.

**Recommended methodological design:**

| Component          | Rigorous Option                                                                          |
|--------------------|------------------------------------------------------------------------------------------|
| Approach           | Sequential mixed (QUAL → QUAN)                                                           |
| Qualitative phase  | Multiple case study (3–5 universities) with in-depth interviews and practice observation |
| Quantitative phase | Survey of teachers and students with validated quality and efficiency scales             |
| Analysis           | Convergent triangulation of both phases                                                  |
| Temporality        | Ideally longitudinal or at least retrospective with historical documentation             |

## 4. AXIOLOGICAL DIMENSION

**What values, interests, and ethical positions traverse this research?**

Central question: From what position of value does the researcher speak, and how do those values silently shape every research decision?

**Axiological tension in the title:**

The title implicitly assumes that Deep Tech-based digital transformation is desirable for university education in emerging contexts. This is a profound axiological position, not a neutral fact, because:

### **1. The technologization of education is politically disputed:**

Adopting Deep Tech in universities of emerging contexts frequently implies dependency on corporate infrastructure (Google, Microsoft, Amazon), whose business models extract data from students and teachers. Studying this transformation without examining these power relations acritically reproduces an agenda that may not coincide with the public educational good.

### **2. The concept of 'emerging contexts' carries implicit values:**

The category 'emerging' places these realities on an evolutionary scale where 'developed' is the desirable horizon — reproducing colonial epistemologies that evaluate Latin American, African, or Southeast Asian education by its distance from Global North models.

### **3. The researcher's positionality matters:**

Does the researcher work at a university adopting these technologies? Do they have ties to technology companies? Are they funded by grants promoting digitalization? Each of these positions generates systematic biases in what is asked, what is measured, and what is concluded.

### **4. Who benefits and who is excluded?**

Technological efficiency in education frequently optimizes for students with stable connectivity, personal devices, and prior digital competencies — reproducing and amplifying pre-existing inequalities in the very emerging contexts the study purports to benefit.

### **Recommended axiological position:**

Explicitly declare:

- The perspective from which the research is conducted and relevant institutional ties
- The values guiding the definition of 'quality' adopted in the study
- A commitment to epistemic justice — including the voices of teachers, students, and communities as knowledge producers, not just research subjects
- Reflection on the ethical limits of recommending technological adoption in contexts where the digital divide can transform 'transformation' into exclusion

## **■ INTEGRATIVE CONCLUSION**

Paradigmatically coherent research on this title cannot limit itself to measuring indicators of technological adoption — it must build from a clear ontology about what educational quality is, an honest epistemology about how it can be known, a methodological design proportional to its claims, and a transparent axiological stance on the values and interests that traverse it.

The coherence between these four dimensions is what distinguishes rigorous research from a superficial technological diagnosis. Paradigmatic coherence is not a bureaucratic requirement —

it is the intellectual integrity that makes any finding trustworthy and any recommendation genuinely useful for the communities this research seeks to serve.